Enhanced Implicit Sentiment Understanding With Prototype Learning and Demonstration for Aspect-Based Sentiment Analysis

Huizhe Su, Xinzhi Wang, Jinpeng Li, Shaorong Xie, and Xiangfeng Luo

***Abstract*—In the field of social computing, the task of aspect-based sentiment analysis (ABSA) aims to classify the sentiment polarity of a given aspect in a sentence. The absence of explicit opinion words in the implicit aspect sentiment expressions poses a greater challenge for capturing their sentiment features in the reviews from social media. Many recent efforts use dependency trees or attention mechanisms to model the association between the aspect and other contextual words. However, dependency tree-based methods are inefficient in constructing valuable associations for sentiment classification due to the lack of explicit opinion words. In addition, the use of attention mechanisms to obtain global semantic information easily leads to an undesired focus on irrelevant words that may have sentiments but are not directly related to the specific aspect. In this article, we propose a novel prototype-based demonstration (PD) model for the ABSA task, which contains prototype learning and PD stages. In the prototype learning stage, we employ mask-aware attention to capture the global sentiment feature of aspect and learn sentiment prototypes through contrastive learning. This allows us to acquire comprehensive central semantics of the sentiment polarity that contains the implicit sentiment features. In the PD stage, to provide explicit guidance for the latent knowledge within the T5 model, we utilize prototypes similar to the aspect sentiment as the neural demonstration. Our model outperforms others with a 1.68%/0.28% accuracy gain on the Laptop/Restaurant datasets, especially in the ISE slice, showing improvements of 1.17%/0.26%. These results confirm the superiority of our PD-ABSA in capturing implicit sentiment and improving classification performance. This provides a solution for implicit sentiment classification in social computing.**

***Index Terms*—Aspect-based sentiment analysis (ABSA), attention mechanism, demonstration, implicit sentiment, prototype, social computing.**

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I. Introduction

SOCIAL computing is the application of computing technology to understand the social context and solve social problems [1]. Sentiment computing is a computational pattern in social computing whose most widespread application is sentiment analysis [1]. Existing sentiment analysis research covers various aspects such as document-level and sentence-level sentiment analysis, as well as multimodal sentiment analysis [2]. Among them, Aspect-based sentiment analysis (ABSA) serves as a fine-grained sentiment analysis task that finds broad application, extending from opinion mining in social computing to customer sentiment analysis in E-commerce platforms [3], [4], [5]. The goal of this task is to classify the sentiment polarity associated with each specified aspect in the sentence, encompassing positive, negative, and neutral sentiments.

Conventional methods focus on capturing sentiment features, which are explicitly expressed by specific words related to the particular aspect. For example, a comment in the Laptop dataset, in Fig. 1, "The new os is great on my macbook pro!," the aspect "os" has a positive sentiment polarity due to the presence of the opinion word "great," which explicitly expresses positive sentiment. While "However I choose two day shipping and it took over a week to arrive," the sentence expresses negative sentiment on the "shipping" aspect. However, there are no explicit words to describe the sentiment of this aspect, making it difficult to classify. And in both the Restaurant and Laptop datasets, similar to the example above, 27.47% and 30.09% of the comments include human-aware implicit sentiment [6]. How to capture the implicit sentiment feature of each aspect is a formidable problem for ABSA.

To address this problem, some previous works [7], [8], [9], [10], [11] exploit dependency trees to model the aspect dependency association with other words in sentences, while others leverage the attention mechanism to extract the connections or relations between the aspect and the other context words. Despite the promising results achieved by these methods, there are still some problems in effectively capturing the implicit sentiment feature of the aspect.

First, the absence of explicit opinion words hinders the use of dependency trees for valuable sentiment associations. Previous approaches [10], [11], [12] analyze the dependency relations between the aspect and the other word within the sentence. Take

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Dataset	Sentence	Implicit	SSEGCN +BERT	Gold label
Laptop	The new os is great on my macbook pro !	False	pos	pos
	However I chose two day shipping and it took over a week to arrive .	True	neu	neg
Restaurant	The Lasagna was probably the best I have tasted .	False	pos	pos
	How can hope to stay in business with service like this ?	True	pos	neg

Fig. 1. Examples of reviews in Laptop and Restaurant datasets.

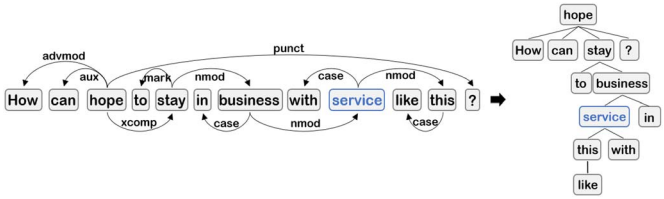


Fig. 2. Dependency relations between the aspect term "service" and other words from the example sentence in the Restaurant dataset.



Fig. 3. Pretrained attention scores between the aspect term "service" and other words in the example sentence from the Restaurant dataset.

the sentence "How can hope to stay in business with service like this?" for example. In this case, there are no explicit sentiment descriptions of the aspect "service," making it impossible to build a direct association for "service" based on the dependency tree for sentiment expression. From the perspective of human understanding, the useful words for classifying the sentiment of the aspect are "can," "hope," "like," "this," and "?." Nevertheless, as illustrated in Fig. 2, the closest nodes to "*service*" in the dependency tree are only "like" and "this," with other words 4 or 5 hops away, making it difficult to associate them with the same attention score to capture contextual semantic information. In particular, "like" usually indicates positive sentiment. In light of this, approaches that rely on dependency trees are inefficient in capturing implicit sentiment context information using the dependency tree structure.

Second, the methods that use the attention mechanism tend to prioritize sentiment words in the absence of explicit opinions, even if they are not aspect-relevant. Traditional attention-based methods [7], [8], [9], [13] learn the aspect-related semantic information with the self-attention. In the sentence "How can hope to stay in business with service like this?," as illustrated in Fig. 3, based on the pretrained attention matrix, the aspect "*service*" exhibits higher attention scores toward words such as "this," "like," "with," "?," and "how." However, these high-scoring words not only fail to assist the model in comprehending the whole semantic information, but also lead the model to overly focus on the word "like," which contains positive sentiment. This results in the model predicting a positive sentiment polarity that is opposite to the gold label. Therefore, traditional attention-based methods are vulnerable to incorporating irrelated sentiment semantic information about the aspect in the sentence.

Inspired by Cui et al. [14], within the same class, the class centroids are expected to encapsulate the implicit semantic features of the class. Thus, the features of each sentiment polarity centroid can be seen as an approximation of that sentiment polarity representation, which is called a prototype. Meanwhile, the prototype representation of each class can serve as a verbalizer, which the model employs to establish a mapping between output words and labels. To capture the implicit sentiment features of the particular aspect, following Cui et al. [14], we use contrastive learning to learn the prototypes of sentiment polarities through the sentiment semantic representation of each aspect within the same polarities. Notably, to tackle the problem of traditional attention including irrelevant sentiment semantic information related to the aspect, we formulate a prompt template for each aspect with a "`[mask]`" element signifying the sentiment features of that aspect. The difference from Cui et al. [14] is that instead of using the "`[mask]`" representation in the input directly for prototype learning, we incorporate global context sentiment information related to the aspect sentiment through a mask-aware attention mechanism prior to the prototype learning. This allows us to obtain the pertinent context information related to the sentiment feature of each aspect hidden in the original sentence to further enrich the aspect's sentiment information.

Furthermore, human beings can exploit associative learning to retrieve related knowledge, thereby augmenting their problem-solving capabilities. Chen et al. [15] retrieves the Top-*m* candidates for each class from a knowledge store constructed with label words found in the training set. The representations of these candidates and label words retrieved from knowledge-store are then incorporated into the encoder for demonstration learning. To leverage the implicit sentiment feature as explicit knowledge, we draw inspiration from this concept. The difference from Chen et al. [15] is that we treat the prototype representations of the three sentiment polarities as a knowledge store, and retrieve prototype representations similar to aspect sentiment. These retrieved representations are then incorporated into the inputs as neural demonstration. This approach leverages implicit sentiment features as a complement to explicit knowledge, enhancing the generalization ability of the model.

In this article, aiming to enhance the capture of implicit aspect sentiment features and utilize these features as explicit knowledge, we introduce a novel prototype demonstration model (PD-ABSA) based on T5 for the ABSA task. Specifically, there are two stages in PD-ABSA, including prototype learning and PD. In the prototype learning stage, we learn the prototypes from the aspect sentiment representation within the same sentiment polarity to capture the implicit sentiment features. To enrich the sentiment features of each aspect, we employ mask-aware attention to obtain the global context information related to the sentiment features of each aspect. In the PD stage, to fully utilize the learned knowledge of the prototypes, we use prototypes similar to the aspect sentiment as neural demonstration. These demonstrations provide explicit

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knowledge guidance for the latent knowledge within the T5 model, enhancing the model's generalization capability.

We evaluate our model's performance on two benchmark datasets and the results demonstrate that PD-ABSA model achieves outstanding performance compared to existing methods. Our contributions can be summarized as follows.

1) We introduce a PD-ABSA model for the ABSA task. This model learns prototypes from aspect sentiment representation to capture implicit sentiment features. Additionally, it utilizes mask-aware attention to capture the relevant contextual information related to the sentiment characteristics of each aspect, thereby enriching the global sentiment feature of each aspect.
2) To fully utilize the knowledge of the prototypes, we use prototypes similar to the aspect sentiment as neural demonstrations to guide T5 with explicit knowledge, enhancing our model's generalization.
3) Experimental results on two benchmark datasets demonstrate the outstanding performance of our PD-ABSA model. Moreover, it is effective in capturing implicit sentiment representations of aspects, which leads to improved classification performance for implicit sentiment.

The subsequent sections in this article are structured as follows. Section II provides a review of related research in the domain of ABSA. Section III presents a concise background of our approach. In Section IV, we dive into a detailed explanation of our methodology. Detailed analysis of the experimental results can be found in Section V. Finally, Section VI offers an all-encompassing summary of our work.

II. Related Work

A. ABSA

ABSA is a fine-grained sentiment analysis task compared to traditional sentiment analysis. Earlier approaches [16], [17] often utilize manually crafted features to predict the sentiment polarity associated with the specific aspects. Subsequently, various methods are proposed for extracting features in sentiment classification, including convolutional and recursive neural network [18], [19], an alignment pretraining framework incorporating instance- and knowledge-level alignments [20], a modularized framework for reproducible ABSA [21], and other approaches.

Recent efforts [10], [11] explicitly incorporate syntactic information from the dependency tree to establish the relationship between aspects and the context. For example, Sun et al. [11] extracts tree-structured syntactic information through graph neutral networks. Liang et al. [22] learns aspect-specific dependencies by constructing a heterogeneous graph. However, these dependency tree-based methods, which link the aspect to the context through the dependency tree, may introduce noise from unrelated relations [23] and fail to address the implicit sentiment problem.

Another trend [7], [8], [9], [24] predominantly utilizes attention mechanisms to model the correlation between the context and the aspects. Among these approaches, Chen et al. [7] categorizes the sentiment polarity of the aspect by introducing a hierarchical attention network. Fan et al. [8] uses a multigrained network to connect the aspects with other words in the sentence. Huang et al. [9] introduces an attention-over-attention (AOA) neural network to model aspects and sentences in a joint manner, explicitly capturing the interaction between aspects and context sentences. Sadr and Solimandarabi [24] aims to leverage both attention mechanisms and transfer learning to enhance sentiment classification performance. Although these attention-based methods are intended to capture implicit semantic relations between the aspect and the context, the attention mechanism sometimes tends to prioritize irrelevant words for the specific aspect sentiment, particularly in the presence of syntactic complexity [25].

In addition, some works combine the syntactic and attention mechanism information to model the association between aspect words and other context words, thereby improving classification performance efficiency. Li et al. [26] combines two types of graph convolutional networks (GCNs), one based on syntax (SynGCN) and the other on semantic correlations (SemGCN), using a self-attention mechanism facilitated by a mutual Bi-Affine module. Zhang et al. [12] learns the aspect-related semantic correlations and constructs syntactic mask matrices to complement the semantic information. However, these methods are not effective in addressing the issue when implicit sentiment expressions (ISEs) lack explicit opinion words.

Furthermore, the pretrained transformer-based models have demonstrated significant success across various NLP tasks, including ABSA. Xu et al. [27] improves the performance on the ABSA task by using BERT [28]. Sun et al. [29] constructs the auxiliary sentences to transform the ABSA task into a sentence pair classification task and subsequently fine-tune the BERT model. Mao et al. [30] treats the ABSA task as a reading comprehension problem, leveraging the pretrained model's knowledge. Zhang et al. [31] formulates the aspect sentiment quad prediction (ASQP) task as a text generation problem to produce the sentiment element sequence. PARAPHRASE [32] models the ASQP task as a paraphrase generation process, making use of the knowledge stored in pretrained models. It is evident that the aforementioned works leverage the latent knowledge in pretrained models for sentiment classification tasks. Some even transform the task into a reading comprehension or generative task to stimulate the latent knowledge of the pretrained model. In this work, we introduce a model that builds upon the T5 model [33] and reframe the ABSA task as a text generation to better exploit the latent knowledge residing within pretrained models.

B. Prompt-Based Tuning

With the birth of GPT-3 [34], much attention has been given to the use of some prompts to stimulate model knowledge [14]. which can help bridge the gap between the masked language model objective and the downstream fine-turning objective. A series of works on exploring knowledge [35] based on prompts has produced impressive results in various tasks. Schick and Schütze [36] designs an essential component of prompt tuning, the pattern verbalizer pair (PVP), which contains a template and a verbalizer. Some works demonstrate

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the effectiveness of the manual verbalizer [36]. However, this approach relies on intensive human labor and expert knowledge. Another research trend is to seek appropriate words from the model's vocabulary, called search-based verbalizer [37], [38]. Nevertheless, the extensive vocabulary poses challenges in finding suitable words. To avoid the above problems, soft verbalizer [39], [40] that uses continuous embeddings as soft labels, attracts more attention. Cui et al. [14] obtains the central points of class instances from few data, called prototypes, to avoid sufficient optimization with a large amount of data.

Similarly, we learn prototype embeddings as the prototype verbalizer from instances within the same sentiment polarity. What's different is that our approach uses a combination of manual verbalizer and prototype verbalizer to improve constraints and retrieves prototypes similar to the aspect sentiment as input for the prototype demonstration.

C. Demonstration

In-context learning (ICL) provides a few prompted instances as part of the input to LLMs without gradient updating [41]. Dynamic retrieval of related training instances for a given input is a common practice for retrieving related demonstrations [41]. To alleviate the limitations of the input length, Chen et al. [15] proposes neural demonstration, which enhances the model's generalization capabilities compared to the discrete demonstration method used in previous work. Unlike Chen et al. [15], we retrieve the prototype embedding similar to aspect sentiment and concatenate the neural demonstration with the input representations after the embedding layer to improve the classification performance through explicit prototype knowledge.

III. Method

In this section, we provide an overview of the background, including task formulation and fine-tuning, then introduce our model. Our model primarily consists of two stages: Prototype learning and PD. Fig. 4 illustrates our model's structure visually.

A. Background

1) Task Formulation: In the conventional ABSA task, an input sentence $X = [x_1, x_2, \ldots, x_{nX}]$ and an aspect term $a \in \{a_1, a_2, \ldots, a_{na}\}$ are provided, where the aspect a is the subsequence of the sentence X. Here, nX represents the length of sentence X and na denotes the number of aspects within the sentence X. The goal is to predict the sentiment polarity ($y \in \mathcal{Y}$) associated with each aspect in the sentence, with $\mathcal{Y} = \{\text{POS,NEG,NEU}\}$ representing the set of sentiment polarities.

2) Fine-Tuning: T5 is a transformer-based [42] model known for its effectiveness in a wide range of natural language processing tasks. We reframe the ABSA task as a text generation problem, employing the T5-base [33] model as the foundational framework. For the standard fine-tuning, the encoder of T5 encode the inputs $X = [x_1, x_2, \ldots, x_{nX}]$ into hidden representation $\mathbf{H}_X = \{\boldsymbol{h}_1, \boldsymbol{h}_2, \ldots, \boldsymbol{h}_{mX}\}$, and the decoder of T5 decode it into a target sequence $Y = [y_1, y_2, \ldots, y_{na}]$, where mX is the length of $\mathbf{H}_X$, $y_i \in \mathcal{Y}$.

B. Prototype Learning

In the prototype learning phase, the goal is to utilize the encoder of T5 to learn the prototypes, specifically, the class centers of the sentiment polarity representations. By achieving class centers that capture the sentiment features of each aspect within the same polarity, we can effectively capture the implicit sentiment features in each sentiment polarity. In the detailed illustration of the prototype learning block in Fig. 4, the dots indicate the sentiment features of the aspect instances, the three colors indicate the three sentiment polarities, and the class center of each sentiment polarity is the prototype feature of that sentiment polarity that we are trying to obtain. In this section, five components are described in detail, including the form of input, the mask-aware attention, prototype learning and the verbalizer.

1) Input: In order to learn the prototype, it is first necessary to obtain the sentiment features of each aspect. We modify the original input X with the template function $\mathcal{T}(\cdot)$ = `emotion of` $< a_i >$: `[mask]`. Each aspect of the sentence is filled in according to this template and spliced together after the sentence using "`[SSEP]`" in the order in which the aspects are placed in the sentence. The input $\hat{X}$ is modified as follows:

$$\begin{aligned}\hat{X} = {} & \text{X [SSEP] emotion of } < a_0 > \text{: [mask] [SSEP] emotion} \\ & \text{of } < a_1 > \text{: [mask] [SSEP]} \ldots \text{ [SSEP] emotion of} \\ & < a_{na} > \text{: [mask]}\end{aligned} \tag{1}$$

where $< a_i >$ is the placeholder for the ith aspect a_i. For example, consider the original sentence: "It does not look like much on the outside, but the minute you walk inside, it's a whole other atmosphere." When analyzing the aspects "*outside*" and "atmosphere," the input is modified as follows: "It does not look like much on the outside, but the minute you walk inside, it's a whole other atmosphere. [SSEP] emotion of outside : [mask] [SSEP] emotion of atmosphere : [mask]."

The encoder of T5 encodes the input $\hat{X}$ into embeddings $\mathbf{H}_{\hat{X}} = \{\boldsymbol{h}_1, \boldsymbol{h}_2, \ldots, \boldsymbol{h}_{m\hat{X}}\}$, where $m\hat{X}$ is the length of $\mathbf{H}_{\hat{X}}$

$$\mathbf{H}_{\hat{X}} = \text{T5Encoder}([\hat{X}]). \tag{2}$$

We extract the hidden state from the last layer of the T5 encoder at the position of the `[mask]` in the input $\boldsymbol{h}_{[\text{mask}]_i}$ as the initial sentiment representation of aspect a_i, where $\boldsymbol{h}_{[\text{mask}]_i}$ is the representation of the ith of `[mask]`, $\boldsymbol{h}_{[\text{mask}]_i} \in \mathbb{R}^d$, d is the dimension of the hidden states. This implementation helps extract the sentiment feature of the aspect while removing irrelevant semantic information.

2) Mask-Aware Attention: After obtaining the initial sentiment features for each aspect, to further enrich these features, we employ mask-aware attention to capture global context information associated with the sentiment features of each aspect in the original sentence. In other words, mask-aware attention employs the multihead attention matrix $\mathbf{A}$ in the T5 encoder to capture the semantic information in the original input X that is related to `[mask]` in the input sentence $\hat{X}$. This enhances our capability to capture implicit sentiment in each sentiment polarity.

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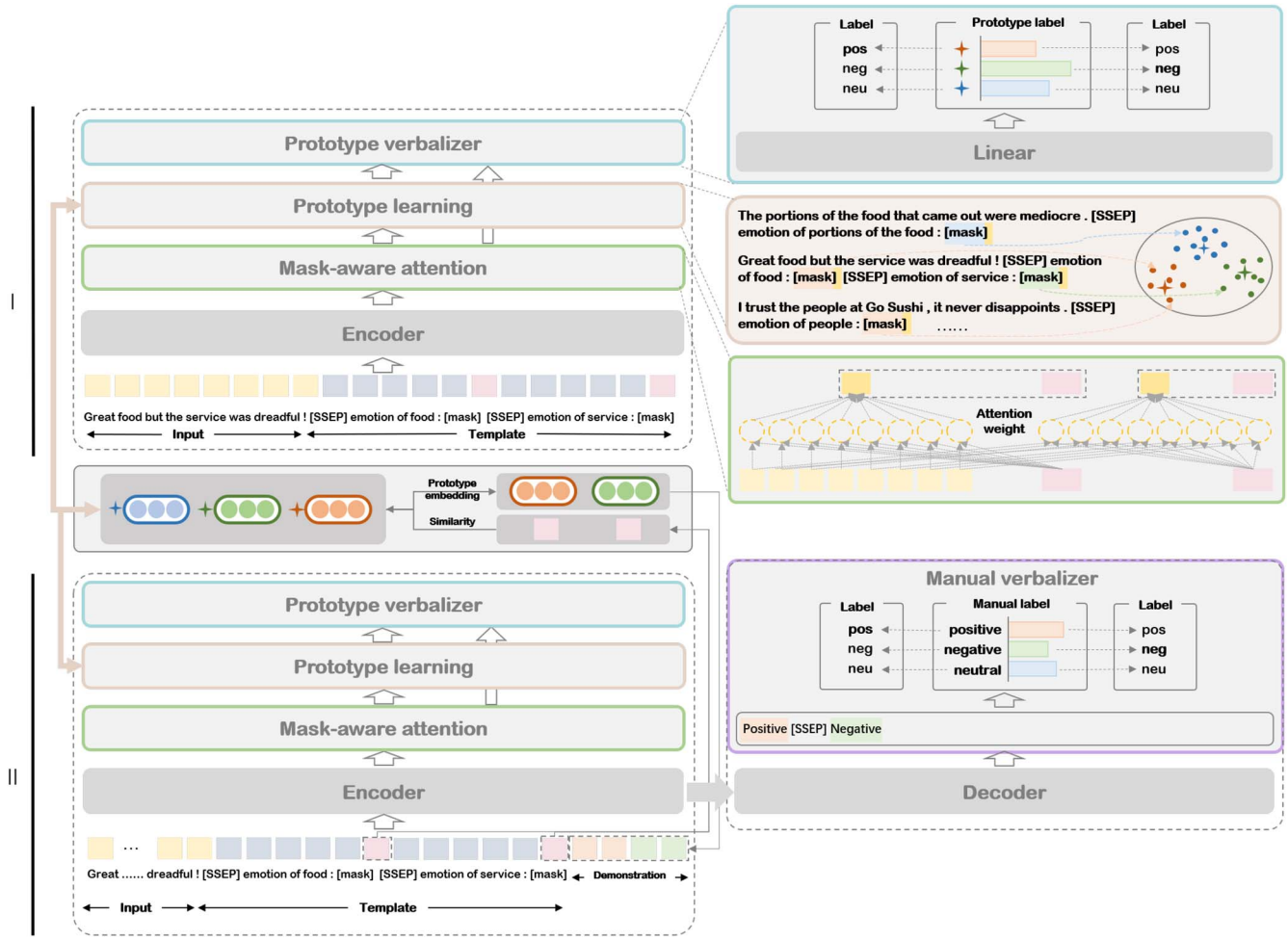


Fig. 4. Overall architecture of our model.

Since the encoder of T5 has already learned token-level dependencies well through multihead self-attention, we leverage these dependencies to acquire global context information about `[mask]` without introducing new attention layers. Given the multihead attention matrix $\mathbf{A} \in \mathbb{R}^{H \times m\hat{X} \times m\hat{X}}$ from the T5 encoder, we extract the attention from the ith `[mask]` to the original input X in the jth attention head as the $\mathbf{A}_{j[\text{mask}]_i X} \in \mathbb{R}^{1 \times mX}$ attention, where $\mathbf{A}_{jkp}$ represents the attention from token k to token p in the jth attention head. After calculating the global context information about the ith `[mask]` in each head, we select the maximum context information among these heads to serve as the final global context information. As a result, the comprehensive sentiment representation of the ith aspect a_i, incorporating the global context information $\mathbf{E}_{a_i}$, is obtained as follows:

$$\mathbf{E}_{a_i} = \max_{j=0}^{H}(\mathbf{A}_{j[\text{mask}]_i X} \cdot \mathbf{H}_{\hat{X}}[1, \ldots, mX]) + \mathbf{h}_{[\text{mask}]_i} \quad (3)$$

where $\mathbf{H}_{\hat{X}}[1, \ldots, mX] \in \mathbb{R}^{mX \times d}$ is the hidden state of the original input tokens in the input $\hat{X}$, $\mathbf{E}_{a_i} \in \mathbb{R}^{1 \times d}$.

3) Prototype Learning: The prototypes are the central points of polarity obtained from the comprehensive aspect sentiment representation of the same polarity. Drawing inspiration from Cui et al. [14], we learn the prototypes through contrastive learning after obtaining the comprehensive sentiment representation for each aspect. Let $\mathcal{C} = \{\boldsymbol{c}_1, \ldots, \boldsymbol{c}_Q\}$ denote the set of prototype representations for sentiment polarities $\mathcal{Y}$, where $\boldsymbol{c}_q \in \mathbb{R}^{1 \times d_{\text{proto}}}$, $Q = 3$ and d_{proto} is the dimension of prototypes. All prototypes in the set $\mathcal{C}$ are randomly initialized before training. To map the ith sentiment representation $\mathbf{E}_{a_i}$ to d_{proto} dimensions, we utilize a linear layer

$$\mathbf{E}_i = W\mathbf{E}_{a_i} \quad (4)$$

where $\mathbf{E}_i \in \mathbb{R}^{1 \times d_{\text{proto}}}$ is the sentiment representation of each aspect a_i. During prototype learning, we utilize the cosine similarity function $S(\cdot)$ to measure the similarity between instances, which are the sentiment representation of aspects $\mathbf{E}_i$

$$S(\mathbf{E}_i, \mathbf{E}_j) = \frac{\mathbf{E}_i}{\|\mathbf{E}_i\|} \cdot \frac{\mathbf{E}_j}{\|\mathbf{E}_j\|}. \quad (5)$$

Following Cui et al. [14], with the instance $\mathbf{E}_i$ and the similarity function $S(\cdot)$, we learn the prototypes by optimizing two objectives. The first objective, *instance-instance*, involves maximizing similarity scores in same polarity while minimizing similarity scores in different polarity between instances; the second objective, *prototype-instance*, aims to maximize similarity scores between the prototypes $\boldsymbol{c}_q$ and the instance

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sharing the same sentiment polarity and minimize similarity scores between the prototypes $\boldsymbol{c}_q$ and the instance with different sentiment polarities. We utilize the following loss functions to optimize the objectives of instance-instance and prototype-instance

$$\mathcal{L}_{\text{ins_ins}} = \frac{-1}{Q^2 B_a^2} \sum_{q} \sum_{i,j} \log \frac{\exp S\left(\mathbf{E}_i^q, \mathbf{E}_j^q\right)}{\sum_{q',j'} \exp S\left(\mathbf{E}_i^q, \mathbf{E}_{j'}^{q'}\right)} \tag{6}$$

$$\mathcal{L}_{\text{proto_ins}} = \frac{-1}{Q^2 B_a} \sum_{i,q} \log \frac{\exp S\left(\mathbf{E}_i^q, \boldsymbol{c}_q\right)}{\sum_{q'} \exp S\left(\mathbf{E}_i^q, \boldsymbol{c}_{q'}\right)} \tag{7}$$

where $(\mathbf{E}_i^q, \mathbf{E}_j^q)$ are the instances in same polarity, the sentiment polarity of $\mathbf{E}_i^q$ is q, B_a is the number of aspects contained in the sentences of a batch. Therefore, the training objective of prototypes $\mathcal{L}_{\text{prototype}}$ is combining the loss of instance-instance and prototype-instance

$$\mathcal{L}_{\text{prototype}} = \mathcal{L}_{\text{ins_ins}} + \mathcal{L}_{\text{proto_ins}}. \tag{8}$$

4) Verbalizer: Prompt-based tuning convert the input into a cloze question, allowing the encoder outputs to be mapped to labels via the Verbalizer. This process bridges the gap between outputs and labels, as discussed in Gao et al. [38]. The verbalizer can be regarded as a projection function that maps sentiment polarity onto words, manually designed or automatically built, or prototypes. Mapping through manually designed words to obtain sentiment labels is called a manual verbalizer, while mapping through sentiment prototypes to obtain sentiment labels is called a prototype verbalizer. In this stage, only sentiment prototypes are selected as verbalizers since the output $\mathbf{E}_i$ are vectors.

We calculate the similarity scores of the query instance and the prototype, with the query instance being the comprehensive sentiment representation $\mathbf{E}_i$ of aspects a_i from the input $\hat{X}$. The probability score of prototype verbalizer for polarity q of the ith aspect sentiment polarity is as follows:

$$P_{\text{proto}}(y_i^q|\hat{X}) = \frac{\exp S\left(\mathbf{E}_i, \boldsymbol{c}_q\right)}{\sum_{q'} \exp S\left(\mathbf{E}_i, \boldsymbol{c}_{q'}\right)} \tag{9}$$

where $P_{\text{proto}}(y_i^q|\hat{X})$ denotes the probability score that the sentiment polarity of the ith aspect a_i in the input $\hat{X}$ is q. The focal loss [43] is used between $P_{\text{proto}}(y_i|\hat{X})$ and the true distribution of the polarity of the ith aspect a_i, summed over all instances as the verbalizer loss $\mathcal{L}_{\text{verbalizer_p}}$ for finetuning the encoder of T5

$$\mathcal{L}_{\text{verbalizer_p}} = -\sum_{i=1}^{K_a} \sum_{q=1}^{Q} \left[\alpha_q \left(1 - P_{\text{proto}}(y_i^q|\hat{X})\right)^{\gamma} \log(P_{\text{proto}}(y_i^q|\hat{X}))\right] \tag{10}$$

where is α_q the polarity weight, γ is the focusing parameter of the focal loss, and K_a is the number of aspects contained in sentences of the training set. Therefore, the final loss of this stage is as follows:

$$\mathcal{L}_{\text{pl}} = \mathcal{L}_{\text{prototype}} + \mathcal{L}_{\text{verbalizer_p}}. \tag{11}$$

Algorithm 1 Prototype Learning

Input: sentence X, aspect a, epochs of stage 1 $epoch_S1$
Output: the set of prototype representations $\boldsymbol{\mathcal{C}}$

```
randomly initialize prototype representations in the set C
for all epoch in epoch_S1 do
    obtain the input X̂ using Eq. 1
    get the embedding of the input H_X̂ using Eq. 2
    obtain the sentiment representation of a_i, denoted as E_{a_i}, using Eq. 3
    update prototype representations in the set C using Eq. 4, Eq. 5, and Eq. 8
    obtain the polarity probabilities of the sentiment label for aspect a_i using prototype verbalizer by Eq. 9
    finetune the Encoder of T5 using Eq. 10 and calculate the final loss using Eq. 11
end for
```

Algorithm 1 presents the algorithm of the Prototype Learning stage. We learn the prototype representation via $\mathcal{L}_{\text{prototype}}$ while fine-tuning the encoder of T5 using $\mathcal{L}_{\text{verbalizer_p}}$ to update the sentiment representation of aspects to better capture the implicit features of sentiment polarity.

C. Prototype-Based Demonstration

After the Prototype Learning stage, the prototype for each sentiment polarity is initially obtained. To enhance the generalization of the model, we propose PDs, where prototypes similar to aspect sentiment are utilized as neural demonstrations, spliced behind the inputs, and fed together into T5. Demonstrations can serve as training examples that guide the model in learning the nuanced relationships between aspects and sentiment. In this way, the prototype demonstrations are utilized to enhance the model's understanding implicit features related to sentiment in the given aspects. In this stage, we give a detailed description of the input and verbalizer components, as they exhibit differences compared to the prototype learning stage.

1) Input: To enhance the model's generalization performance using prototypes, we concatenate neural demonstrations with the input $\hat{X}$ at the embedding layer following Chen et al. [15]. First, we obtain the input representation $\mathbf{Z}_{\hat{X}} \in \mathbb{R}^{mX \times d}$ at the embedding layer and utilize the representation $\mathbf{Z}_{[\text{mask}]_i} \in \mathbb{R}^{1 \times d}$ of the ith `[mask]` as the initial sentiment representation of the ith aspect a_i

$$\mathbf{Z}_{\hat{X}} = e(\hat{X}) \tag{12}$$

$$\mathbf{Z}_i = W \mathbf{Z}_{[\text{mask}]_i} \tag{13}$$

where $e(\cdot)$ represents the word embedding layer of T5 model. Then, we use $\mathbf{Z}_i \in \mathbb{R}^{1 \times d_{\text{proto}}}$ to query the representations of prototypes to calculate the similarity scores, where W is the trainable weights. Thus, we obtain the prototype representation

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with the highest similarity score, denoted as $\boldsymbol{c}_i^*$

$$y_i^* = \arg\max_q \frac{\exp S\left(\mathbf{Z}_i, \boldsymbol{c}_q\right)}{\sum_{q'} \exp S\left(\mathbf{Z}_i, \boldsymbol{c}_{q'}\right)} \tag{14}$$

$$\boldsymbol{c}_i^* = W_c \boldsymbol{c}_{y_i^*} \tag{15}$$

where y_i^* represents the polarity within the sentiment polarity set $\mathcal{Y}$ for the ith aspect a_i. $\boldsymbol{c}_{y_i^*} \in \mathbb{R}^{1\times d_{\text{proto}}}$ represents the representations of prototypes for the sentiment polarity y_i^*. After linear transformation, $\boldsymbol{c}_i^* \in \mathbb{R}^{1\times d}$ is the representations of prototypes for the sentiment polarity y_i^* with different dimension, where W_c is the trainable weights.

The prototypes $\boldsymbol{c}_i^*$ most similar to the sentiment representation of each aspect are selected and then spliced after the input representation $\mathbf{Z}_{\hat{X}}$ as demonstrations to enhance the model's understanding of implicit features related to sentiment in the given aspects

$$\boldsymbol{\mathcal{I}} = \mathbf{Z}_{\hat{X}} \oplus \boldsymbol{c}_1^* \oplus \cdots \oplus \boldsymbol{c}_i^* \oplus \cdots \oplus \boldsymbol{c}_{na}^* \tag{16}$$

where the symbol $\oplus$ indicates the concatenation operation, and $\boldsymbol{\mathcal{I}}$ denotes the input features that are fed into the T5.

Subsequently, the comprehensive sentiment representation of the ith aspect a_i is similarly obtained using (2) and (3), as in stage 1. The learning of the prototype continues using (4), (5), and (8).

2) Verbalizer: In the stage, we employ a mix verbalizer, including both the prototype verbalizer and the manual verbalizer. Specifically, prototypes and manual labels are used as verbalizers, respectively. This differs from the prototype learning stage.

In the T5 encoder, we compute the probability score for the polarity q of the prototype verbalizer according to the (9), which is the same as in the prototype learning stage. In the T5 decoder, the manually designed label set $\mathcal{V} = \{\text{positive,negative,neutral}\}$ is used as the manual verbalizer. In this setup, manual label words for each sentiment polarity are not the primary focus of our research. Therefore, we define only one vocabulary for each sentiment polarity. The projection function that maps the sentiment polarity in sentiment polarity set $\mathcal{Y}$ to the manual label word is as follows:

$$f(y_i) = \begin{cases} \text{positive}, & \text{if } y_i = \text{POS} \\ \text{negative}, & \text{if } y_i = \text{NEG} \\ \text{neutral}, & \text{if } y_i = \text{NEU}. \end{cases} \tag{17}$$

Since the sentiment of each aspect is expressed by the corresponding manually designed label word $v \in \mathcal{V}$. The target sequence Y, which is the output sequence of the T5 decoder, is constructed by concatenating the sentiment expression word of each aspect with the special symbol `[SSEP]`. The probability score of the manual verbalizer for the polarity q of the ith aspect sentiment polarity is as follows:

$$P_{\text{manual}}(y_i^q|\boldsymbol{\mathcal{I}}) = \frac{1}{N_q} \sum_{j=1}^{N_q} \frac{P_{\mathcal{M}}(f(y_i))_j^q - \mu}{\sigma} \tag{18}$$

where $P_{\mathcal{M}}(f(y_i))_j^q$ denotes the logit of the jth manual label word of polarity q for the ith aspect, and $\mathcal{M}$ represents the T5 model. Here, μ represents the mean of these logits, σ stands for the standard deviation of the logits, and N_q is the number of manual label words for the qth sentiment polarity ($N_q = 1$). Consequently, the probability score of mix verbalizer for the polarity q of the ith aspect sentiment polarity is as follows:

$$P_{\text{mix}}(y_i^q|\boldsymbol{\mathcal{I}}) = \frac{P_{\text{proto}}(y_i^q|\boldsymbol{\mathcal{I}}) + P_{\text{manual}}(y_i^q|\boldsymbol{\mathcal{I}})}{2} \tag{19}$$

$$\mathcal{L}_{\text{verbalizer_m}} = -\sum_{i=1}^{K_a}\sum_{q=1}^{Q} \left[\alpha_q \left(1 - P_{\text{mix}}(y_i^q|\boldsymbol{\mathcal{I}})\right)^{\gamma} \log(P_{\text{mix}}(y_i^q|\boldsymbol{\mathcal{I}}))\right]. \tag{20}$$

Similarly, we use the focal loss as the verbalizer loss, and fine-tuning T5 in this stage using $\mathcal{L}_{\text{verbalizer_m}}$ and $\mathcal{L}_{\text{LLM}}$. So, the final loss of this stage is as follows:

$$\mathcal{L}_{\text{pd}} = \mathcal{L}_{\text{LLM}} + \mathcal{L}_{\text{verbalizer_m}} + \mathcal{L}_{\text{prototype}} \tag{21}$$

where L_{LLM} denote the large language modeling loss.

Algorithm 2 presents the algorithm of the PD stage. The prototype demonstration is used as explicit knowledge to guide T5 while continuing to learn to update the prototype.

During inference, we only utilize the PD model with the trained prototypes to predict the sentiment polarity of each aspect. This approach offers the similar prototype as supplementary knowledge, rather than relying solely on the knowledge hidden in the parameters of the T5 model.

Algorithm 2 Prototype-based Demonstration

Input: sentence X, aspect a, epochs of stage 2 $epoch_S2$, the set of prototype representations $\boldsymbol{\mathcal{C}}$ learned in the first stage

Output: the set of prototype representations $\boldsymbol{\mathcal{C}}$, the fine-tuned T5 model

1: **for all** epoch in epoch_S2 **do**
2: get the input $\hat{X}$ using Eq. 1
3: obtain the input representation $Z_{\hat{X}}$ at the embedding layer using Eq. 12
4: get the initial sentiment representation of a_i, denoted as Z_i, using Eq. 13
5: obtain the similar prototypes of sentiment representation of a_i using Eq. 14 and Eq. 15
6: get the input of T5 $\boldsymbol{\mathcal{I}}$ by splicing the similar prototypes of a_i after the input representation $Z_{\hat{X}}$ as the demonstration, as shown in Eq. 16
7: after feeding the input $\boldsymbol{\mathcal{I}}$ to T5, the comprehensive sentiment representation of the i-th aspect a_i is similarly obtained using Eq. 2 and Eq. 3
8: continue to update prototype representations in the set $\boldsymbol{\mathcal{C}}$ using Eq. 4, Eq. 5, and Eq. 8
9: obtain the polarity probabilities of the sentiment label for aspect a_i using mix verbalizer by Eq. 19
10: finetune the T5 using by $\mathcal{L}_{LLM}$ and Eq. 20, and calculate the final loss by Eq. 21
11: **end for**

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TABLE I
STATISTICS OF THE DATASETS

Datasets	Division	Positive	Negative	Neutral	Total
Laptop	Train	976	851	455	2282
	Test	337	128	167	632
	Total	1313	979	622	2914
Restaurant	Train	2164	807	637	3608
	Test	727	196	196	1119
	Total	2891	1003	833	4727
Twitter	Train	1507	1528	3016	6051
	Test	172	169	336	677
	Total	1679	1697	3352	6728

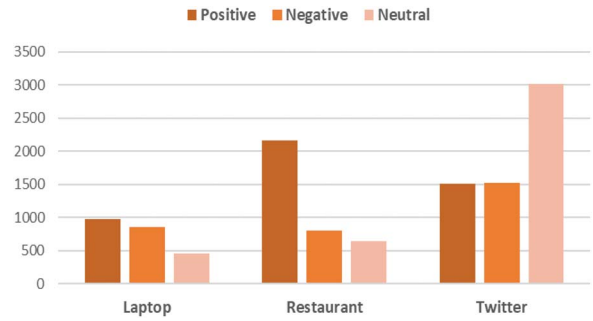


Fig. 5. Distribution of sentiment labels in the training set of the three datasets.

IV. EXPERIMENTS

A. Datasets

We primarily perform experiments on three benchmark datasets for ABSA, including the Laptop and Restaurant reviews from SemEval 2014 Task 4 [44], and Twitter from Dong et al. [45]. The Laptop and Restaurant datasets contain manually annotated reviews of laptops and restaurants [44]. The Twitter dataset consists mainly of twitter, including news, user's views on events, sentiments shared by users', and anything posted by users on social media [46]. Specifically, we adopt the three datasets provided by Zhang et al. [12]. The sentiment categories of these datasets are divided into positive, negative and neutral. The dataset statistics are provided in Table I. Fig. 5 visually presents the sentiment distribution in the training set of the three datasets. In the Laptop dataset, there is a slight imbalance between the number of positive and negative sentiments, with less data available for neutral sentiments. In the Restaurant dataset, positive sentiment data is the most abundant, while both negative and neutral sentiment data are less than positive sentiment data, with little difference between the two. In the Twitter dataset, neutral sentiment is the most prevalent, and positive and negative sentiment are nearly equal, each comprising half the amount of neutral sentiment data.

In addition, to validate the performance of our model in capturing the implicit sentiment, we added implicit annotations for each aspect from Li et al. [6] in the Laptop and Restaurant dataset test set to the corresponding aspect of each sentence in Zhang et al. [12]. Finally, we manually checked the implicit sentiment label of each aspect to ensure correctness. To ensure fairness of the comparison, we strictly adhered to the dataset configuration used in the previous study. The final test set with added implicit sentiment labels is consistent with the number of Zhang et al. [12], and the implicit label statistics of the Laptop and Restaurant dataset test set are shown in Table II.

TABLE II
IMPLICIT LABEL STATISTICS OF THE LAPTOP AND RESTAURANT DATASET TEST SET

Datasets	Division	Positive	Negative	Neutral	Total
Laptop-test	Explicit	301	100	59	460
	Implicit	36	28	108	172
	Total	337	128	167	632
Restaurant-test	Explicit	652	142	59	853
	Implicit	75	54	137	266
	Total	727	196	196	1119

TABLE III
HYPERPARAMETERS IN TRAINING

Parameter	Laptop	Restaurant	Twitter
Epoch	35	35	20
Batchsize	20	22	22
lr	5e–5	5e–5	2e–5
Weight decay	1e–3	5e–4	5e–4
Epoch of I	20	20	5
Epoch of II	15	15	15

B. Implementation Details

In our implementation, we build our framework using the T5-base model available from the Hugging Face Transformers library. The T5-base model consists of 12 layers in both the encoder and decoder, with 12 attention heads. We set the seed of the model to 42. The hidden state dimension of the T5-base model is 768, whereas we set the prototype dimension to be 64 and randomly initialize it at the start of training. For the focal loss, α_q and γ are set to 0.25 and 2, respectively. We train our model utilizing the Adam optimizer with epsilon (eps) values set at 1e–8, respectively. The optimizer's learning rate (lr), weight decay, and batchsize vary across datasets, with detailed settings shown in Table III.

C. Baseline Methods

Dependency tree-based and attention-based methods face limitations in capturing the implicit sentiment of aspects. Our model is designed to address these challenges. In order to fully evaluate the performance of our model in capturing implicit sentiment features, we conduct a comparative analysis with the attention-based methods, the dependency tree-based methods, and the hybrid methods. Details of the models used for comparison are given as follows:

1) Attention-based methods.
   a) *BERT-SPC* [47] models association between context and target aspect using attention scores based on BERT encoder.

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b) *R-GAT + BERT* [25] utilizes a relational graph attention model to capture the aspect-oriented tree structure and establish connections between aspects and opinions.
c) *BERTAsp* [6] captures aspect-related sentiment information by applying aspect-aware fine-tuning on BERT-base.
d) *ACDC + BERT* [48] extracts and integrates deep semantic features from the aspects and contexts at different levels using a dense concatenation mechanism.

2) Dependency tree-based methods.
a) *RepWalk* [49] focus on the crucial context information by conducting a replicated random walk on a dependency graph.
b) *TGCN + BERT* [50] proposes a type-aware GCN that takes into account both word relations and dependency types to learn from the results of dependency parsing.
c) *LSOIT* [46] builds a knowledge-aware opinion induction tree that integrates the lexicon and syntax knowledge through reinforcement learning and attention mechanisms.

3) Combination of the two methods.
a) *MTABSA* [51] exploits multihead attention to associate dependency sequences with aspects to improve the connection between aspects and their associated dependency sequences by combining Aspect Polarity Classification (APC) and Aspect Term Extraction (ATE) tasks.
b) *dotGCN* [52] proposes an aspect-specific and language-agnostic discrete latent opinion tree model, and applies reinforcement learning and attention mechanisms to extract aspect-specific features.
c) *DualGCN + BERT* [26] employs a BiAffine module to connect the SynGCN module, responsible for capturing syntax knowledge, with the SemGCN module, which focuses on semantic correlations. Furthermore, two regularizers are designed to boost the performance of these modules.
d) *SSEGCN + BERT* [12] proposes an aspect-aware attention with self-attention to learn semantic correlations from aspect-related to global and utilizes syntactic mask matrices to supplement with semantic information.

To thoroughly assess the effectiveness of our model, we introduce the following variants as baselines.

a) *T5:* We perform direct fine-tuning on the T5-base, which serves as the base model for PD-ABSA. The input is constructed by concatenating each aspect after the original sentence using the `[SSEP]` symbol, following the order of appearance in the sentence. For example, "It does not look like much on the outside, but the minute you walk inside, it's a whole other atmosphere. [SSEP] outside [SSEP] atmosphere." The target output is the sentiment polarity of each aspect, utilizing the T5 vocabulary.
b) *T5 + prompt*: We also fine-tune on the T5-base, which is the same as "T5." The difference is that the input is to fill each aspect into the template "`emotion of {} : [mask],`" which is spliced after the original sentence using the `[SSEP]` symbol. For example, "It does not look like much on the outside, but the minute you walk inside, it's a whole other atmosphere. [SSEP] emotion of outside : [mask] [SSEP] emotion of atmosphere : [mask]."
c) *T5 + prompt + prototype*: This variant adds prototype learning to "T5 + prompt," which includes the mask-aware attention and the mix verbalizer, and the other settings are the same as "T5 + prompt."

D. Evaluation Metrics

We evaluate the performance of our model and compare it with the baselines using accuracy (in percent) and macro-averaged F1-score (in percent) as the evaluation metrics, denoted as Acc and Macro-F1, respectively. In addition, to assess the performance regarding explicit and implicit sentiment, and following the evaluation metrics used in Li et al. [6], we partition the test set into explicit sentiment expression (ESE) and ISE slices based on whether the sentiment of the aspect is explicit or implicit. We report the accuracy (in percent) on the ESE slices and ISE slices as the evaluation metrics, denoted by ESE and ISE, respectively. Furthermore, to evaluate the performance regarding different polarities, we report the micro-averaged F1-score (in percent) on the positive, negative, and neutral polarities, denoted as POS, NEG, and NEU, respectively.

E. Main Results

1) The Performance of the Baselines and Our Model: The performance of the baselines and our model on the datasets is reported in Table IV. The combined methods outperform most of the attention-based methods and the dependency tree-based methods. This phenomenon suggests that both capturing the association between aspect and opinion words through the attention mechanism and learning the syntactic structure of sentences are effective strategies for ABSA.

Our model achieves higher accuracy and ISE (accuracy on ISE slices) compared to all the baseline models on the Laptop and Restaurant datasets. This improvement is observed across attention-based methods, dependency tree-based methods, and combinations of both. Compared to attention-based methods, our PD-ABSA method achieves a performance gain of 1.68%/0.28% on the Laptop/Restaurant datasets in terms of accuracy, and outperforms dependency tree-based methods by 0.48%/0.49%. Notably, upon reimplementing the DualGCN + BERT and SSEGCN + BERT models, we observe that our model surpasses even the combined methods on the Laptop and Restaurant datasets, particularly in terms of accuracy on the ISE slice, with improvements of 1.17%/0.26%, respectively. Additionally, our model achieves an accuracy on ESE slices that is 1.15% higher than in the Restaurant dataset. These results suggest that, although our model does not utilize the syntactic knowledge of sentences, it can effectively capture the implicit sentiment features of aspects in the sentence by learning prototypes and enhance implicit sentiment by combining

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TABLE IV
MAIN RESULTS OF DIFFERENT METHODS ON DATASETS

Method		Laptop				Restaurant				Twitter	
		Acc	Macro-F1	ESE	ISE	Acc	Macro-F1	ESE	ISE	Acc	Macro-F1
Attention-based	BERT-SPC	78.22	73.45	81.47	69.54	83.57	77.16	89.21	65.54	73.55	72.14
	R-GAT + BERT	78.21	74.07	82.44	72.99	86.60	81.35	92.73	67.79	76.15	74.88
	BERTAsp	78.53	74.07	82.33	68.39	85.80	78.95	92.73	63.67	–	–
	ACDC + BERT	80.41	76.56	–	–	87.09	81.41	–	–	76.37	74.68
Dependency tree-based	RepWalk	78.20	76.90	–	–	83.80	76.90	–	–	74.40	72.60
	TGCN + BERT	80.88	77.03	–	–	86.16	79.95	–	–	76.45	75.25
	LSOIT	81.41	77.16	–	–	86.88	**82.27**	–	–	77.75	76.79
Combination	MTABSA	80.56	77.00	–	–	86.88	81.16	–	–	76.59	74.67
	dotGCN	81.03	78.10	–	–	86.16	80.49	–	–	**78.11**	**77.00**
	DualGCN + BERT	81.80	78.10	–	–	87.13	81.16	–	–	77.40	76.02
	DualGCN + BERT*	81.80	78.43	83.47	77.32	86.33	80.34	92.02	68.04	–	–
	SSEGCN + BERT	81.01	77.96	–	–	87.31	81.09	–	–	77.40	76.02
	SSEGCN + BERT*	79.91	76.48	**84.13**	68.60	86.42	80.97	92.03	68.42	–	–
	Our PD-ABSA	**82.09**	**78.45**	83.44	**78.49**	**87.37**	81.34	**93.18**	**68.68**	75.18	74.51

Note: ESE and ISE denote accuracy on explicit sentiment expression slices and implicit sentiment expression slices of the Laptop and Restaurant datasets, respectively. The baseline methods marked with * are the result of our reimplementation. The highest performance is highlighted in bold.

TABLE V
MAIN RESULTS OF DIFFERENT LARGE LANGUAGE MODELS ON DATASETS

Datasets	Models	Macro-F1	Acc	ESE	ISE	POS	NEG	NEU
Laptop	LLaMa2-7b	0.60	0.68	0.76	0.47	0.81	0.54	0.44
	LLaMa2-13b	0.65	0.73	0.82	0.48	0.85	0.65	0.46
	ChatGPT-3.5	0.73	0.78	**0.85**	0.59	0.87	**0.77**	0.55
	Our PD-ABSA	**0.78**	**0.82**	0.83	**0.78**	**0.91**	0.76	**0.69**
Restaurant	LLaMa2-7b	0.63	0.76	0.84	0.50	0.87	0.69	0.35
	LLaMa2-13b	0.67	0.78	0.86	0.53	0.89	0.73	0.38
	ChatGPT-3.5	0.68	0.81	0.89	0.54	0.90	0.78	0.36
	Our PD-ABSA	**0.81**	**0.87**	**0.93**	**0.68**	**0.93**	**0.84**	**0.66**
Twitter	LLaMa2-7b	0.44	0.42	–	–	0.46	0.42	0.46
	LLaMa2-13b	0.41	0.41	–	–	0.49	0.35	0.41
	ChatGPT-3.5	0.56	0.55	–	–	0.57	0.64	0.46
	Our PD-ABSA	**0.75**	**0.74**	–	–	**0.71**	**0.75**	**0.77**

Note: The evaluation metrics used are accuracy, accuracy on the ESE and ISE slices, and micro-averaged F1-score in different sentiment polarity. Note that the scores are not expressed as percentages. The highest performance is highlighted in bold.

explicit prototype knowledge with the information hidden in the PLM parameters.

We also notice that our model achieves slightly lower accuracy on ESE slices compared to the DualGCN + BERT and SSEGCN + BERT model approaches in the Laptop dataset. The statistics in Table II reveal that the Laptop test set comprises 72% explicit sentiment data, which is lower than the Restaurant dataset, accounting for 76% explicit sentiment data. This may result in the prototype learning insufficient explicit sentiment features, leading to slightly suboptimal results on ESE.

For the Twitter dataset, our model does not achieve optimal results, as shown in Table IV. Additionally, based on the results in Table V, the LLMs ChatGPT-3.5 does not achieve high F1 scores for all three sentiment polarities on this dataset. These observations suggest that LLMs' semantic of the three sentiments in Twitter data is prone to confusion, making it challenging to effectively distinguish them. Therefore, employing T5 to encode Twitter, obtaining three sentiment features with challenging semantic distinctions, and subsequently learning prototypes cannot effectively capture the implicit features of each sentiment. Nevertheless, as shown in Table V, our model's F1 score for neutral sentiment in the Twitter dataset is 0.77, which is the highest compared to the scores for neutral sentiment in the Laptop and Restaurant datasets. In Fig. 5, neutral sentiment data in Twitter constitutes almost half of the dataset. Therefore, this could be attributed to the abundance of neutral sentiment data, enabling our model to learn implicit features of neutral sentiments. This compensates for the semantic confusion in language models.

2) The Performance of the LLMs and Our Model: To further assess the performance of the LLMs, we utilize the llama2-7b-chat-v2 and llama2-13b-chat-v2 APIs provided by

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TABLE VI
EXPERIMENTAL RESULTS FROM ABLATION STUDY OF DIFFERENT COMPONENTS OF THE PD-ABSA ON THE LAPTOP AND RESTAURANT DATASETS

Method	Laptop				Restaurant			
	Acc	Macro-F1	ESE	ISE	Acc	Macro-F1	ESE	ISE
T5	74.96	69.24	78.65	65.12	81.63	72.93	87.54	62.64
T5 + prompt	76.55	71.74	79.52	68.60	85.48	77.93	92.60	62.64
T5 + prompt + prototype	80.82	77.37	81.92	77.91	86.29	78.84	92.48	66.42
T5 + prompt + prototype + demonstration(PD-ABSA)	**82.09**	**78.45**	**83.44**	**78.49**	**87.37**	**81.34**	**93.18**	**68.68**
PD-ABSA w/o mask-aware attention	80.19	76.65	81.05	77.91	85.30	76.34	93.30	59.62

Note: The highest performance is highlighted in bold.

AliCloud's Lingji Platform,[1] along with the ChatGPT-3.5 API provided by OpenAI[2] to conduct tests on three datasets.

The results of the three LLMs are presented in Table V. As the number of parameters in the LLMs increases, all the scores show improvement in the Laptop and Restaurant datasets. However, in the Twitter dataset, the scores of Macro-F1, ACC, NEG, and NEU for llama-7b are superior to those for llama-13b. The three LLMs exhibit significantly lower performance in classifying positive and negative sentiment on the Twitter dataset compared to the Laptop and Restaurant datasets. This further suggests the challenge in distinguishing between positive and negative sentiments on the Twitter dataset. While each of the three LLMs performs well, especially ChatGPT-3.5, our method surpasses ChatGPT-3.5 in Macro-F1 and ACC by 5%/4%, 13%/6%, and 19%/19% on the Laptop, Restaurant, and Twitter datasets, respectively. On the ISE slices, our method outperforms the Laptop, Restaurant dataset by 19%,14% respectively, highlighting its effectiveness in capturing implicit sentiment features through prototype learning and demonstration. Moreover, on the Twitter dataset, our method achieves F1 scores for positive, negative and neutral sentiments that outperform ChatGPT-3.5 by 14%/11%/31%, respectively. This indicates that our method excels in learning sentiment features across different polarities compared to these LLMs, especially in scenarios where distinguishing between positive and negative sentiment is challenging.

F. Ablation Study

1) Effect of Different Components of PD-ABSA: To further explore the effectiveness of individual components in our model, we perform ablation studies. Among the variants listed in Section IV-C, the difference between "T5 + prompt" and "T5" lies in the input. In the former case, the input involves splicing phrases after the original sentence, where the phrases are generated by aspect-filled templates, whereas in the latter case, the aspects are spliced into the sentence. As illustrated in Table VI, the results of "T5 + prompt" are consistently superior to "T5" in the Laptop and Restaurant datasets, confirming the effectiveness of the prompt components of PD-ABSA.

"T5 + prompt + prototype" differs from "T5 + prompt" in that it incorporates the learning of prototypes, as well as including mask-aware attention and the mix verbalizer. The rest of the setup remains the same as "T5 + prompt." All results of "T5 + prompt + prototype" demonstrate substantial improvements, with gains of 4.27%/0.81% and 9.31%/3.78% in term of accuracy and accuracy on the ISE slice for the Laptop and Restaurant, respectively. This verifies that learning prototypes of sentiment polarity to obtain the implicit sentiment features and capturing the global context information by mask-aware attention to supplement sentiment features of each aspect are essential for PD-ABSA.

In addition, compared to "PD-ABSA," "T5 + prompt + prototype" removes the prototype similar to aspect sentiment as a demonstration, which significantly decreases the model performance. The accuracy decreases by 1.27%/1.08% on the Laptop and Restaurant datasets, particularly on the ISE slices, the accuracy decreases by 0.58%/2.26%. This observation not only underscores prototype-based presentations can further enhance the capture of implicit features, but also suggests that the essential role of prototype-based presentations in augmenting explicit prototype knowledge with the pretrained large model, thus contributing significantly to the effectiveness of PD-ABSA.

Furthermore, when we exclude the mask-aware attention from the "PD-ABSA" model, there is a noticeable decline in performance, with accuracy dropping by 1.9%/2.07%. This underscores the significance of mask-aware attention in capturing the correlated global context of each aspect. Notably, when mask-aware attention is removed, there is a decrease of 0.85% and 9.3% in accuracy on the ISE slice for the Laptop and Restaurant datasets. This further suggests that, in the Restaurant dataset, there are more sentences than in the Laptop dataset that need to be taken into account for the global context information of each aspect.

Comparing all the variants listed in Section IV-C, we find that each module is indispensable. Prototype learning proved to be the most effective in improving accuracy, followed by prototype examples. In addition, mask-aware attention had the greatest impact on capturing implicit emotional features in the Restaurant dataset. This observation also suggests that the Restaurant dataset has more data on implicit sentiment features in the global context than the Laptop dataset.

[1]https://help.aliyun.com/zh/dashscope/developer-reference/llama2-big-language-model-fine-tune

[2]https://platform.openai.com/

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TABLE VII
EXPERIMENTAL RESULTS FROM ABLATION STUDY OF DIFFERENT KINDS OF VERBALIZERS WITHIN THE PD-ABSA FRAMEWORK

Method	Laptop					Restaurant				
	Acc	ISE	POS	NEG	NEU	Acc	ISE	POS	NEG	NEU
PD-ABSA_proto	0.80	0.76	0.90	0.75	0.65	**0.87**	**0.68**	**0.93**	**0.84**	**0.66**
PD-ABSA_manual	0.78	0.72	0.89	0.72	0.63	0.83	0.56	0.92	0.80	0.54
PD-ABSA_mix	**0.82**	**0.78**	**0.91**	**0.76**	**0.69**	0.86	0.65	**0.93**	**0.84**	0.62

Note: The evaluation metrics used are accuracy, accuracy on the ISE slice, and micro-averaged F1-score in different sentiment polarity. Note that the scores are not expressed as percentages. The highest performance is highlighted in bold.

TABLE VIII
EXPERIMENTAL RESULTS FROM ABLATION STUDY OF VARIOUS PROMPT TEMPLATES IN THE PD-ABSA

Template	Laptop				Restaurant			
	Acc	Macro-F1	ESE	ISE	Acc	Macro-F1	ESE	ISE
Emotion of {} : [mask]	**82.09**	**78.45**	**83.44**	78.49	**87.37**	**81.34**	**93.18**	**68.68**
The [mask] emotion : {}	81.30	77.71	83.01	76.74	86.38	79.30	92.83	65.66
{} : [mask]	80.98	77.97	81.70	**79.07**	84.50	76.53	92.24	59.62
The emotion of {} means [mask]	80.67	77.43	81.70	77.91	84.59	75.32	92.01	60.76
The emotion of aspect : {} means [mask]	80.82	77.33	81.70	78.49	83.78	73.75	92.13	56.98

Note: The highest performance is highlighted in bold.

2) Effect of Different Kinds of Verbalizers Within the PD-ABSA: To demonstrate the effectiveness of prototype verbalizer, manual verbalizer and mix verbalizer in PD-ABSA, we compare our model using different verbalizers and report the results in Table VII. We observe that in the Laptop dataset, the results of the mix verbalizer are 4%/6% better than those of the manual verbalizer, and 2%/2% better than those of the prototype verbalizer in terms of accuracy and accuracy on the ISE slice, respectively. Additionally, the mix verbalizer achieves the highest scores in terms of F1 scores for different sentiment polarity. This phenomenon suggests that the prototypes used by prototype verbalizer effectively represent the sentiment polarity. Meanwhile, the words chosen by the manual verbalizer accurately convey the semantic information of the three sentiment polarities in this dataset.

While in the Restaurant dataset, the prototype verbalizer demonstrates the highest performance, surpassing the manual verbalizer and the mix verbalizer by 4%/12% and 1%/3%, respectively in terms of accuracy and accuracy on the ISE slice. For the POS and NEG polarities, both the prototype verbalizer and the mix verbalizer yield the same F1 scores. Notably, in the NEU polarity, the F1 score of the prototype verbalizer surpasses that of the mix verbalizer by 4%, illustrating that the manually labeled neutral words in the manual verbalizer fail to accurately describe the neutral sentiment semantic information in this dataset.

3) Effect of Various Prompt Templates in the PD-ABSA: We further perform an ablation study to evaluate the efficacy of different prompt templates in PD-ABSA and the results are as shown in Table VIII. These templates take into account the formats of previous works and are designed by intuition. We find that there can be significant differences in results when using different prompt templates.

The first template "`emotion of {} : [mask]`" performs better than other templates, although the accuracy on the ISE slice in the Laptop dataset is lower than the third template "`{} : [mask].`" Among these templates, the fifth template only has "`aspect:`" more than the fourth template, showing that even a small change in the templates can make a difference. Similarly, the results of the first two templates reveal that an inconsistent order of template tokens can also affect the final results. Moreover, the third one "`{} : [mask]`" performs unsatisfactorily compared to the first prompt template, possibly because the template is not concise and lacks useful prompt information for the large pretrained model.

The results indicate that nuances in the templates can also have an impact, as the templates should not be overly complex, which can incorporate noisy information into the model, or overly succinct, which results in a model that lacks useful cues.

G. Analysis

1) Effectiveness of the Prototype Representation: In order to gain a deeper understanding of the effectiveness of the prototype representation, we employ PCA [53] to reduce the dimensionality of the aspect sentiment representation and perform visualization, as illustrated in Fig. 6. Fig. 6(a) illustrates the sentiment polarity representation of each aspect in the "T5 + prompt" model without prototype learning. We observe that the sentiment representations of each aspect with the same sentiment polarity are scattered and not converging. On the other hand, Fig. 6(b) showcases the sentiment polarity representation of each aspect in the PD-ABSA model. The visualization reveals that the PD-ABSA model, with the prototype learning module, effectively clusters sentiment representations of aspects with the same sentiment polarity, capturing both implicit and explicit sentiment features.

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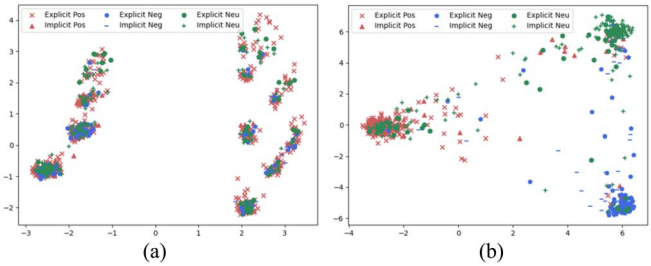


Fig. 6. Visualization of hidden sentiment representation on Restaurant dataset. (a) T5 + prompt. (b) PD-ABSA.

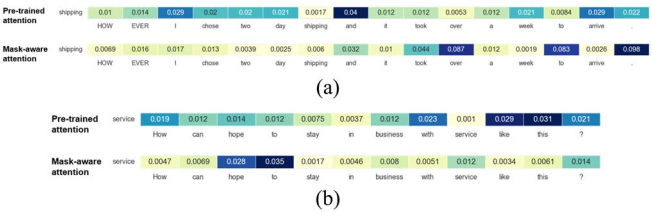


Fig. 7. Example visualization of attention scores of mask-aware in PD-ABSA model on the Laptop and Restaurant datasets. (a) Example in Laptop dataset. (b) Example in Restaurant dataset.

However, we also observed that the sentiment representations of neutral polarity are misclassified as positive or negative polarity. Combining the observation with the F1 scores in different sentiment polarities shown in Table VII, the fact that F1 scores in neutral polarity are lower than scores in positive and negative polarities suggests that capturing neutral sentiment features is indeed challenging, and that prototype learned based on aspect sentiment expressions do not completely alleviate this issue.

2) Effectiveness of the Mask-Aware Attention in PD-ABSA: To further demonstrate the effectiveness of the mask-aware attention in PD-ABSA, we provide visualization of attention scores of two examples that are incorrectly classified in other models, as shown in Fig. 7. For the example from the Laptop dataset in Fig. 7(a), "However I chose two day shipping and it took over a week to arrive.." The self-attention mechanism in the T5 encoder focuses on almost all tokens, without particularly focusing on those tokens that could have a significant impact on the conveyed sentiment, such as "over" and "to." In contrast, the mask-aware attention in our model is capable of focusing on words that are contribute to the overall semantics, even if these words are distant from the aspect "shipping." Notably, these words that aspect focuses on in this sentence, such as "over," "to," and even the punctuation "." can provide the neutral sentiment expression with negative sentiment semantics, thereby assisting in correct classification.

Similarly, for the sentence in the Restaurant dataset in Fig. 7(b), "How can hope to stay in business with service like this ?." The self-attention mechanism in the T5 encoder pays significant attention to words such as "like," "this," and even focuses on words other than the aspect, including "?," but as "hope" and "like" convey positive semantics, this could lead to a wrong judgment. In contrast, the mask-aware attention mechanism in our model places emphasis not only on "hope," but also on "can," "this" and the punctuation "?." These words assist our model in comprehensively capturing contextual information related to aspects, enabling it to uncover negative sentiment semantics related to aspects and accurately identify sentiment polarity.

Thus, the mask-aware attention in PD-ABSA can capture aspect-related contextual semantics that are useful for correct classification of sentiment polarity.

Sentence	Gold label	T5	SSEGCN +BERT	DualGCN +BERT	Ours
However I chose two day shipping and it took over a week to arrive .	neg	neu ✗	neu ✗	neg ✓	neg ✓
Can you buy any laptop that matches the quality of a MacBook ?	pos	neu ✗	neu ✗	pos ✓	pos ✓
It doesn't look like much on the outside , but the minute you walk inside , it's a whole other atmosphere .	neg, pos	pos, ✗ neu ✗	neg, ✓ pos ✓	neg, ✓ pos ✓	neg, ✓ pos ✓
How can hope to stay in business with service like this ?	neg	pos ✗	pos ✗	pos ✗	neg ✓

Fig. 8. Case studies of our model compared with several baselines.

H. Case Study

We perform a case study to explore whether PD-ABSA can effectively classify the sentiment polarity of an aspect with ISE, as demonstrated in Fig. 8. In particular, we compare PD-ABSA with "T5," SSEGCN + BERT, DualGCN + BERT, showing their predicted sentiment polarity for the aspect that highlighted in the blue font. It is evident that our PD-ABSA model surpasses all other baseline models. For the aspect "shipping" of the sentence "However I chose two day shipping and it took over a week to arrive," the "T5" and the SSEGCN methods fail to capture the implicit negative semantic information and incorrectly predict neutral sentiment polarity. In contrast, our model correctly classifies the sentiment polarity as negative, thanks to the combination of prototype learning and PD.

Moreover, for the aspect "service" of the hard case, "How can hope to stay in business with service like this ?," all the baselines predicted the sentiment polarity as positive which may be misleading because of the positive sentiment word "hope." Our model can capture the global semantics without interference from some words to correctly eliminate the negative sentiment polarity, which proves the effectiveness of implicit sentiment representation of aspects of our PD-ABSA model.

In order to examine whether the ability of different variants of our model to mine implicit sentiment in twitter can provide data support for social computing applications, such as analyzing a group's emotional attitudes toward a topic, we conduct experiments using cases from twitter about COVID-19. As shown as in Fig. 9, We can observe that although none of the four twitter expresses a clear opinion, the relevant sentiment words in twitter 3 and 4 are more explicit, such as "provide better protection" and "no evidence...being safe." Therefore, the different variants of our model yield correct results for both twitter 3 and 4. However, in the case of 1 and 2, where the sentiment polarity is ambiguous and difficult to judge, the results of the "T5," "T5 + prompt + prototype," and "PD-ABSA w/o mask-aware attention" variants are all incorrect.

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Num	Sentence	Gold label	T5	T5+prompt+ verbalizer	PD-ABSA w/o mask-aware attention	Ours
1	New Scientist @newscientist · Sep 9 Children between the ages of 4 months and 2 years who have had covid-19 are more likely to have antibodies that attack insulin-producing cells, a feature of type 1 diabetes	neu	neg ✗	pos ✗	neg ✗	neu ✓
2	Data Report @CovidDataReport · 12h University of Maryland students living on campus will need to isolate at their permanent homes or at a hotel room for five days if they test positive for COVID-19.	neu	neg ✗	-- ✗	neg ✗	neu ✓
3	U.S. FDA @US_FDA · 13h Today, we took action approving & authorizing for emergency use updated COVID-19 vaccines formulated to more closely target currently circulating variants & to provide better protection against serious consequences of COVID. Here's what you need to know: fda.gov/news-events/pr...	pos	pos ✓	pos ✓	pos ✓	pos ✓
4	Dr. Kat Lindley @KLVeritas · 13h Florida's Surgeon General, Dr. Joseph Ladapo, warned that there was "no evidence" of the upcoming COVID-19 vaccines being safe for human beings and suggested that people may be better off not getting these new jabs.	neg	neg ✓	neg ✓	neg ✓	neg ✓

Fig. 9. Results of experimental comparisons among different variants of our model on the case of COVID-19 from Twitter.

Only our model, "PD-ABSA," is able to accurately identify the sentiments expressed in the twitter, which suggests that either of the modules in our model is important for classifying sentiment polarity, and not a single one can be omitted.

The cases mentioned above serve as strong evidence of our model's superior performance in mining implicit sentiment and accurately capturing sentiment features, even in situations where sentiment polarity in social media comments is ambiguous. This opens up greater possibilities for sentiment computing, social data analysis in the field of social computing.

V. Conclusion

In this article, we present PD-ABSA, an innovative PD model for the ABSA task, designed for application in social computing. Our model comprises two stages: Prototype learning captures sentiment polarity features, particularly implicit sentiments, while PD uses aspect-sentiment prototypes to guide T5's latent knowledge explicitly. Experimental results confirm not only the superiority of our PD-ABSA model over existing models on Laptop and Restaurant datasets but also its effectiveness in capturing the implicit sentiment representation of aspects and improving the classification performance of implicit sentiment. Further experiments reveal that while our model outperforms ChatGPT-3.5 on Twitter, it does not achieve optimal results. This could be attributed to T5, such as LLMs, struggling with semantic confusion among the three sentiments in Twitter data, and our model facing difficulty in capturing implicit sentiment features with semantically hard-to-distinguish sentiments. Therefore, we identify this challenge as a subject for future research, and we are committed to exploring effective solutions.

References

[1] W. Wang, H. Ning, F. Shi, S. Dhelim, W. Zhang, and L. Chen, "A survey of hybrid human-artificial intelligence for social computing," *IEEE Trans. Human-Mach. Syst.*, vol. 52, no. 3, pp. 468–480, Jun. 2022.

[2] L. Zhu, Z. Zhu, C. Zhang, Y. Xu, and X. Kong, "Multimodal sentiment analysis based on fusion methods: A survey," *Inf. Fusion*, vol. 95, pp. 306–325, 2023.

[3] W. Zhang, X. Li, Y. Deng, L. Bing, and W. Lam, "A survey on aspect-based sentiment analysis: Tasks, methods, and challenges," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 11, pp. 11019–11038, Nov. 2023.

[4] Z. Khodaverdian, H. Sadr, S. A. Edalatpanah, and M. Nazari, "An energy aware resource allocation based on combination of CNN and GRU for virtual machine selection," *Multimedia Tools Appl.*, vol. 83, no. 9, pp. 25769–25796, 2023.

[5] S. Soleymanpour, H. Sadr, and M. N. Solimandarabi, "CSCNN: Cost-sensitive convolutional neural network for encrypted traffic classification," *Neural Process. Lett.*, vol. 53, no. 5, pp. 3497–3523, 2021.

[6] Z. Li, Y. Zou, C. Zhang, Q. Zhang, and Z. Wei, "Learning implicit sentiment in aspect-based sentiment analysis with supervised contrastive pre-training," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2021, pp. 246–256.

[7] P. Chen, Z. Sun, L. Bing, and W. Yang, "Recurrent attention network on memory for aspect sentiment analysis," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2017, pp. 452–461.

[8] F. Fan, Y. Feng, and D. Zhao, "Multi-grained attention network for aspect-level sentiment classification," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2018, pp. 3433–3442.

[9] B. Huang, Y. Ou, and K. M. Carley, "Aspect level sentiment classification with attention-over-attention neural networks," in *Proc. 11th Int. Conf. Social, Cultural, Behav. Model.*, vol. 10899, 2018, pp. 197–206.

[10] C. Zhang, Q. Li, and D. Song, "Aspect-based sentiment classification with aspect-specific graph convolutional networks," in *Proc. Conf. Empirical Methods Natural Lang. Process/9th Int. Joint Conf. Natural Lang. Process.*, 2019, pp. 4567–4577.

[11] K. Sun, R. Zhang, S. Mensah, Y. Mao, and X. Liu, "Aspect-level sentiment analysis via convolution over dependency tree," in *Proc. Conf. Empirical Methods Natural Lang. Process/9th Int. Joint Conf. Natural Lang. Process.*, 2019, pp. 5678–5687.

[12] Z. Zhang, Z. Zhou, and Y. Wang, "SSEGCN: Syntactic and semantic enhanced graph convolutional network for aspect-based sentiment analysis," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol.*, 2022, pp. 4916–4925.

[13] Y. Wang, M. Huang, X. Zhu, and L. Zhao, "Attention-based LSTM for aspect-level sentiment classification," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2016, pp. 606–615.

[14] G. Cui, S. Hu, N. Ding, L. Huang, and Z. Liu, "Prototypical verbalizer for prompt-based few-shot tuning," in *Proc. 60th Annu. Meeting Assoc. Comput. Linguistics* (Volume 1 Long Papers), 2022, pp. 7014–7024.

[15] X. Chen et al., "Decoupling knowledge from memorization: Retrieval-augmented prompt learning," in *Proc. Annu. Conf. Neural Inf. Process. Syst.*, 2022.

Authorized licensed use limited to: UNIVERSITAS TELKOM. Downloaded on April 15,2026 at 10:45:24 UTC from IEEE Xplore. Restrictions apply.

[16] L. Jiang, M. Yu, M. Zhou, X. Liu, and T. Zhao, "Target-dependent Twitter sentiment classification," in *Proc. 49th Annu. Meeting Assoc. Comput. Linguistics: Human Lang. Technol., Proc. Conf.*, 2011, pp. 151–160.
[17] S. Kiritchenko, X. Zhu, C. Cherry, and S. M. Mohammad, "NRC-Canada-2014: Detecting aspects and sentiment in customer reviews," in *Proc. 8th Int. Workshop Semantic Eval.*, 2014, pp. 437–442.
[18] H. Sadr, M. M. Pedram, and M. Teshnehlab, "A robust sentiment analysis method based on sequential combination of convolutional and recursive neural networks," *Neural Process. Lett.*, vol. 50, no. 3, pp. 2745–2761, 2019.
[19] H. Sadr, M. M. Pedram, and M. Teshnehlab, "Multi-view deep network: A deep model based on learning features from heterogeneous neural networks for sentiment analysis," *IEEE Access*, vol. 8, pp. 86984–86997, 2020.
[20] H. Yang, C. Zhang, and K. Li, "PyABSA: A modularized framework for reproducible aspect-based sentiment analysis," in *Proc. 32nd ACM Int. Conf. Inf. Knowl. Manage.*, 2023, pp. 5117–5122.
[21] J. Liu, Q. Zhong, L. Ding, H. Jin, B. Du, and D. Tao, "Unified instance and knowledge alignment pretraining for aspect-based sentiment analysis," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 31, pp. 2629–2642, 2023.
[22] B. Liang, R. Yin, L. Gui, J. Du, and R. Xu, "Jointly learning aspect-focused and inter-aspect relations with graph convolutional networks for aspect sentiment analysis," in *Proc. 28th Int. Conf. Comput. Linguistics*, 2020, pp. 150–161.
[23] S. Liang, W. Wei, X. Mao, F. Wang, and Z. He, "BiSyn-GAT+: Bi-Syntax aware graph attention network for aspect-based sentiment analysis," in *Proc. Findings Assoc. Comput. Linguistics*, 2022, pp. 1835–1848.
[24] H. Sadr and M. N. Solimandarabi, "ACNN-TL: Attention-based convolutional neural network coupling with transfer learning and contextualized word representation for enhancing the performance of sentiment classification," *J. Supercomputing*, vol. 78, no. 7, pp. 10149–10175, 2022.
[25] K. Wang, W. Shen, Y. Yang, X. Quan, and R. Wang, "Relational graph attention network for aspect-based sentiment analysis," in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics*, 2020, pp. 3229–3238.
[26] R. Li, H. Chen, F. Feng, Z. Ma, X. Wang, and E. H. Hovy, "Dual graph convolutional networks for aspect-based sentiment analysis," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics 11th Int. Joint Conf. Natural Lang. Process.*, 2021, pp. 6319–6329.
[27] H. Xu, B. Liu, L. Shu, and P. S. Yu, "BERT post-training for review reading comprehension and aspect-based sentiment analysis," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol.*, 2019, pp. 2324–2335.
[28] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol.*, 2019, pp. 4171–4186.
[29] C. Sun, L. Huang, and X. Qiu, "Utilizing BERT for aspect-based sentiment analysis via constructing auxiliary sentence," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol.*, 2019, pp. 380–385.
[30] Y. Mao, Y. Shen, C. Yu, and L. Cai, "A joint training dual-MRC framework for aspect based sentiment analysis," in *Proc. 35th AAAI Conf. Artif. Intell.*, 2021, pp. 13543–13551.
[31] W. Zhang, X. Li, Y. Deng, L. Bing, and W. Lam, "Towards generative aspect-based sentiment analysis," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics/11th Int. Joint Conf. Natural Lang. Process.*, 2021, pp. 504–510.
[32] W. Zhang, Y. Deng, X. Li, Y. Yuan, L. Bing, and W. Lam, "Aspect sentiment quad prediction as paraphrase generation," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2021, pp. 9209–9219.
[33] C. Raffel et al., "Exploring the limits of transfer learning with a unified text-to-text transformer," *J. Mach. Learn. Res.*, vol. 21, no. 10, pp. 140:1–140:67, 2020.
[34] T. B. Brown et al., "Language models are few-shot learners," in *Proc. Annu. Conf. Neural Inf. Process. Syst.*, 2020, pp. 1877–1901.
[35] J. Davison, J. Feldman, and A. M. Rush, "Commonsense knowledge mining from pretrained models," in *Proc. Conf. Empirical Methods Natural Lang. Process/9th Int. Joint Conf. Natural Lang. Process.*, 2019, pp. 1173–1178.
[36] T. Schick and H. Schütze, "Exploiting cloze-questions for few-shot text classification and natural language inference," in *Proc. 16th Conf. Eur. Chapter Assoc. Comput. Linguistics*, 2021, pp. 255–269.
[37] T. Shin, Y. Razeghi, R. L. L. IV, E. Wallace, and S. Singh, "AutoPrompt: Eliciting knowledge from language models with automatically generated prompts," in *Proc. Conf. Empirical Methods Natural Lang. Process.*, 2020, pp. 4222–4235.
[38] T. Gao, A. Fisch, and D. Chen, "Making pre-trained language models better few-shot learners," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics/11th Int. Joint Conf. Natural Lang. Process.*, 2021, pp. 3816–3830.
[39] K. Hambardzumyan, H. Khachatrian, and J. May, "WARP: Word-level adversarial reprogramming," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics/11th Int. Joint Conf. Natural Lang. Process.*, 2021, pp. 4921–4933.
[40] N. Zhang et al., "Differentiable prompt makes pre-trained language models better few-shot learners," in *Proc. 10th Int. Conf. Learn. Representations*, 2022, pp. 1–19.
[41] Z. Zhang, A. Zhang, M. Li, and A. Smola, "Automatic chain of thought prompting in large language models," in *Proc. 11th Int. Conf. Learn. Representations*, 2023, pp. 1–32.
[42] A. Vaswani et al., "Attention is all you need," in *Proc. Annu. Conf. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.
[43] T. Lin, P. Goyal, R. B. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, early access, 2017.
[44] M. Pontiki, D. Galanis, J. Pavlopoulos, H. Papageorgiou, I. Androutsopoulos, and S. Manandhar, "SemEval-2014 task 4: Aspect based sentiment analysis," in *Proc. 8th Int. Workshop Semantic Eval.*, 2014, pp. 27–35.
[45] L. Dong, F. Wei, C. Tan, D. Tang, M. Zhou, and K. Xu, "Adaptive recursive neural network for target-dependent twitter sentiment classification," in *Proc. Annu. Meeting Assoc. Comput. Linguistics*, 2014, pp. 49–54.
[46] H. Wu, D. Zhou, C. Sun, Z. Zhang, Y. Ding, and Y. Chen, "LSOIT: Lexicon and syntax enhanced opinion induction tree for aspect-based sentiment analysis," *Expert Syst. Appl.*, vol. 235, 2024, Art. no. 121137.
[47] Y. Song, J. Wang, T. Jiang, Z. Liu, and Y. Rao, "Attentional encoder network for targeted sentiment classification," 2019, *arXiv:1902.09314*.
[48] G. Ma and X. Guo, "Dense concatenation memory network for aspect level sentiment analysis," *IEEE Access*, vol. 11, pp. 20486–20493, 2023.
[49] Y. Zheng, R. Zhang, S. Mensah, and Y. Mao, "Replicate, walk, and stop on syntax: An effective neural network model for aspect-level sentiment classification," in *Proc. -34th AAAI Conf. Artif. Intell.*, 2020, pp. 9685–9692.
[50] Y. Tian, G. Chen, and Y. Song, "Aspect-based sentiment analysis with type-aware graph convolutional networks and layer ensemble," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Lang. Technol.*, 2021, pp. 2910–2922.
[51] G. Zhao, Y. Luo, Q. Chen, and X. Qian, "Aspect-based sentiment analysis via multitask learning for online reviews," *Knowl. Based Syst.*, vol. 264, 2023, Art. no. 110326.
[52] C. Chen, Z. Teng, Z. Wang, and Y. Zhang, "Discrete opinion tree induction for aspect-based sentiment analysis," in *Proc. Annu. Meeting Assoc. Comput. Linguistics*, 2022, pp. 2051–2064.
[53] M. E. Tipping and C. M. Bishop, "Mixtures of probabilistic principal component analysers," *Neural Comput.*, vol. 11, no. 2, pp. 443–482, 1999.



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